Masters

Parallel and Distributed Computing (CS621)

Question: 1 (Marks: 1) Attempted Questions: 0 Total Questions: 50

In _____ control strategy, the control function is distributed among different components in the system.

Choices:

- decentralized
- distributed
- centralized
- supervised

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Parallel and Distributed Computing (CS621)

Question: 2 (Marks: 1) Attempted Questions: 1 Total Questions: 50

In _____ switching, a complete path has to be established prior to the start of communication between a source and a destination.

Choices:

- packet
- port
- circuit
- connect

Masters

Parallel and Distributed Computing (CS621)

Question: 3 (Marks: 1) Attempted Questions: 2 Total Questions: 50

Number of bits transmitted in a unit of time is known as_____.

Choices:

- latency
- diameter
- bandwidth ✓
- bisection

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Parallel and Distributed Computing (CS621)

Question: 4 (Marks: 1) **Attempted Questions: 3** **Total Questions: 50**

A _____ topology is a network interconnection scheme in which each node is connected to every other node with a direct link.

Choices:

- bus
- star
- mesh ✓
- switch

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Parallel and Distributed Computing (CS621)

Question: 5 (Marks: 1) Attempted Questions: 4 Total Questions: 50

The possibility to use the same software in different environments is known as _____.

Choices:

- reliability
- portability
- robustness
- availability

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Parallel and Distributed Computing (CS621)

Question: 10 (Marks: 1) **Attempted Questions: 6** **Total Questions: 50**

Decomposition of computation into a large number of tasks results in _____ decomposition.

Choices:

- fine-grained
- course-grained
- medium-grained
- best-grained

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Parallel and Distributed Computing (CS621)

Question: 11 (Marks: 1) Attempted Questions: 7 Total Questions: 50

Discrete optimization problems can be solved by using _____ decomposition technique.

Choices:

- Recursive decomposition
- Data decomposition
- Exploratory decomposition
- Speculative decomposition

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Parallel and Distributed Computing (CS621)

Question: 12 (Marks: 1) Attempted Questions: 8 Total Questions: 50

The problem of finding the minimum in a list is an example of decomposition of data based on _____

Choices:

- output data
- input data
- intermediate data
- both input and output data

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Parallel and Distributed Computing (CS621)

Question: 13 (Marks: 1) Attempted Questions: 9 Total Questions: 50

Which of the following decomposition techniques is/are more desirable for quicksort problem?

Choices:

- Recursive decomposition
- Data decomposition
- Exploratory decomposition
- Both data and recursive decomposition

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Parallel and Distributed Computing (CS621)

Question: 26 (Marks: 1) **Attempted Questions: 10** **Total Questions: 50**

Sequential Consistency ensures that the result of any execution in a distributed system is equivalent to:

Choices:

- Operations being executed in a random order
- All processes executing concurrently without synchronization
- Read and write operations being executed in a sequential order
- Each process executing independently of others

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Parallel and Distributed Computing (CS621)

Question: 27 (Marks: 1) Attempted Questions: 11 Total Questions: 50

What does causal consistency guarantee regarding writes in a distributed system?

Choices:

- All writes are seen by all processes in the same order.
- Concurrent writes are seen in the same order on different machines.
- All writes are executed concurrently to improve performance.
- Writes that are causally related are seen in the same order by all processes.

Parallel and Distributed Computing (CS621)

Question: 29 (Marks: 1) **Attempted Questions: 13** **Total Questions: 50**

GPUs have gained popularity due to their ability to:

Choices:

- Run only real-world applications efficiently.
- Execute single-threaded tasks faster than multi-core CPUs.
- Replace multi-core CPUs in all computing tasks.
- Provide parallel processing capabilities for certain applications. ✓

Parallel and Distributed Computing (CS621)

Question: 30 (Marks: 1) **Attempted Questions: 14** **Total Questions: 50**

CUDA is designed for computation-intensive tasks to be executed on which hardware?

Choices:

- CPU (Central Processing Unit)
- GPU (Graphics Processing Unit)
- RAM (Random Access Memory)
- SSD (Solid State Drive)

Parallel and Distributed Computing (CS621)

Question: 31 (Marks: 1) Attempted Questions: 15 Total Questions: 50

GPUs accelerated libraries are commonly used in which type of applications?

Choices:

- Text editors
- Web browsers
- Data science and machine learning
- File compression utilities

Parallel and Distributed Computing (CS621)

Question: 32 (Marks: 1) Attempted Qu

In message passing, how do processors exchange data in a distributed memory system?

Choices:

- They share a global memory space.
- They use cache synchronization mechanisms.
- They communicate through a communication network.
- They directly access each other's local memory.

Question: 34 (Marks: 1) Attempted Questions: 18 Total Questions: 50

What is the size of the Basic MPI library in terms of the number of functions?

Choices:

- Large (over 50 functions)
- Medium (around 25 functions)
- Small (around 10 functions)
- Basic (6 functions)



Parallel and Distributed Computing (CS621)

Question: 35 (Marks: 1) Attempted Questions: 19 Total Questions: 50

In MPI, which function is commonly used to send a message from one process to another?

Choices:

- MPI_Broadcast
- MPI_Reduce
- MPI_Send
- MPI_Receive

Parallel and Distributed Computing (CS621)

Question: 36 (Marks: 1) Attempted Questions: 20 Total Questions: 50

MapReduce is a software framework which supports _____computing on large data sets.

Choices:

- serial
- parallel
- distributed
- both parallel and distributed

Parallel and Distributed Computing (CS621)

Question: 37 (Marks: 1) Attempted Questions: 21 Total Questions: 50

Which of the following is NOT true regarding Google File System (GFS)?

Choices:

- High security
- Limited security
- Only viable in a specific environment
- Custom designed

Parallel and Distributed Computing (CS621)

Question: 38 (Marks: 1) Attempted Questions: 22 Total Questions: 50

_____ of Hadoop Distributed File System (HDFS) focuses on the individual instances of application data or the data content.

Choices:

- Structural metadata
- Descriptive metadata
- Contextual metadata
- Contextual and structural metadata

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 39 (Marks: 1) **Attempted Questions: 23** **Total Questions: 50**

What is the primary function of Hadoop's HDFS (Hadoop Distributed File System)?

Choices:

- Running complex machine learning algorithms.
- Storing and distributing data across a cluster of machines. ✓
- Managing user access control.
- Creating graphical visualizations

Parallel and Distributed Computing (CS621)

Question: 40 (Marks: 1) Attempted Questions: 24 Total Questions: 50

Which component of Hadoop is responsible for storing and managing data across a cluster?

Choices:

- Hadoop Mapper
- Hadoop Reducer
- Hadoop NameNode
- Hadoop TaskTracker

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 41 (Marks: 3) **Attempted Questions: 25** **Total Questions: 50**

Define the degree of the switch in interconnection networks.

.....

Answer:

The total number of ports on a switch is the degree of the switch.
The cost of a switch grows as the square of the degree of the switch, the peripheral hardware linearly as the degree, and the packaging costs linearly as the number of pins.

Parallel and Distributed Computing (CS621)

Question: 42 (Marks: 3) Attempted Questions: 26 Total Questions: 50

Bill Dally, a prominent computer scientist has highlighted power efficiency and massive parallelism as major advantages of GPUs by considering two aspects. You are required to mention these aspects.

Answer:

Ans:Two Aspects:
1.Data Access Rate Capability
2.Data Processing Capability

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 43 (Marks: 3) **Attempted Questions: 28** **Total Questions:**

Write down the names of the data-centric consistency models.

.....

Answer:

Data-centric Consistency Models Types Names:

I. Continuous Consistency
II. Consistent Ordering of Operations
III. Causal Consistency
IV. Grouping Operations

Parallel and Distributed Computing (CS621)

Question: 44 (Marks: 3) Attempted Questions: 28 Total Questions: 50

What will happen when multiple threads attempt to manipulate the same data item, if proper care is not taken to synchronize?

.....

Answer:

Can results incoherent

Parallel and Distributed Computing (CS621)

Question: 45 (Marks: 3) Attempted Questions: 29 Total Questions: 5(

The I/O tools are the collections of system software and libraries have grownup to address I/O issues.
You are required to list three I/O tools.

Answer:

Here is the names of three I/O tools.

Collections of system software and libraries have grownup to address I/O issues:
- At Parallel file systems
- MPI-IO • High level libraries
- Relationships between these are not always clear. Choosing between tools can be diffcult.

Parallel and Distributed Computing (CS621)

Question: 46 (Marks: 3) Attempted Questions: 30 Total Questions:

Do you think, a parallel runtime of a program can be evaluated asymptotically as a function of input size only? Justify your answer in either case.

Answer:

Analytical Modeling – Basics
The parallel runtime of a program depends on the input size, the number of processors, and the communication parameters of the machine.
A sequential algorithm is evaluated by its runtime (in general, asymptotic runtime as a function of input size)

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 47 (Marks: 5) **Attempted Questions: 31**

Write down any five interconnection network's properties in parallel and distributed computing.

.....

Answer:

Following are five interconnection network's properties in parallel and distributed computing:

five properties of interconnection networks in parallel and distributed computing. Here are five key properties:

Topology: The structure or arrangement of nodes and links, such as mesh, torus, or hypercube. It determines how nodes are interconnected.

Latency: The time it takes for a message to travel from the source to the destination.

Bandwidth: The amount of data that can be transmitted over a link in a given amount of time.

Scalability: The ability of the network to handle an increasing number of nodes without significant performance degradation.

Fault Tolerance: The capability of the network to continue functioning correctly even when some components fail.

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 49 (Marks: 5) **Attempted Questions: 34** **Total Questions: 5**

The process of dividing a computation into smaller parts, some or all of which may potentially be executed in parallel, is called decomposition. You are required to list 4 key decomposition techniques.

.....

Answer:

Following are the 4 key decomposition techniques.

Decomposition Techniques
- Recursive Decomposition
- Data Decomposition
- Exploratory Decomposition
- Speculative Decomposition

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 50 (Marks: 5) **Attempted Questions: 34** **Total Questions: 50**

Differentiate the following statement as Independent MPI IO and Collective MPI IO calls.

1. MPI_File_write()
2. MPI_File_write_all()
3. MPI_File_read().
4. MPI_File_read_all()

Answer:

Independent MPI-IO
MPI_File_write()
MPI_File_read().

Collective MPI-IO:
MPI_File_write_all()
MPI_File_read_all()

Parallel and Distributed Computing (CS621)

Question: 7 (Marks: 1) Attempted Questions: 35 Total Questions: 50

Much of the effort associated with writing correct threaded programs is spent on _____ concurrent threads.

Choices:

- scheduling
- data accesses for
- synchronizing
- both scheduling and data access

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 8 **(Marks: 1)** **Attempted Questions: 36** **Total Questions: 50**

Which of the following can be used to overcome the overhead of polling for availability of locks?

Choices:

- Semaphores
- Mutex-Locks
- Condition variable
- trylock () function

Parallel and Distributed Computing (CS621)

Question: 9 (Marks: 1) Attempted Questions: 37 Total Questions: 50

The length of the _____in a task dependency graph is called the critical path length.

Choices:

- shortest path
- longest path
- average of longest and shortest path
- intersection of longest and shortest path

Parallel and Distributed Computing (CS621)

Question: 13 (Marks: 1) Attempted Questions: 38

Which of the following decomposition techniques is/are more desirable for quicksort problem?

Choices:

- (●) Recursive decomposition
- ( ) Data decomposition
- ( ) Exploratory decomposition
- Both data and recursive decomposition

Parallel and Distributed Computing (CS621)

Question: 14 (Marks: 1) Attempted Questions: 38 Total Questions: 50

Keep in view of Lustre file system, Identify, which of the following provides file service, and network request handling for one or more local OSTs?

Choices:

- Matadata Server (MDS)
- Matadata Target (MDT)
- Object Storage Server (OSS)
- Object Storage Target (OST)

Parallel and Distributed Computing (CS621)

Question: 15 (Marks: 1) Attempted Questions: 39 Total Questions: 5(

Keep in view of Lustre file system, Identify, which of the following stores file data (chunks of files)?

Choices:

- Matadata Server (MDS)
- Matadata Target (MDT)
- Object Storage Server (OSS)
- Object Storage Target (OST)



Parallel and Distributed Computing (CS621)

Question: 16 (Marks: 1) Attempted Questions: 40 Total Questions: 5

Which of the following is not true regarding POSIX I/O interface?

Choices:

- Standard I/O interface across many platforms
- open, read/write, close functions in C/C++/Fortran
- Mechanism almost all serial applications use to perform I/O
- Many ways for describing collective access

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 17 (Marks: 1)

Which of the following is not true regarding MPI-IO?

Choices:

- Provides a high-level interface
- Has a large number of routines
- Facilitate Collective I/O
- Facilitate Atomicity rules

Parallel and Distributed Computing (CS621)

Question: 18 (Marks: 1) **Attempted Questions: 42** **Total Questions: 50**

Aggregates smaller read/write operations into larger operations is an example of_____.

Choices:

- POSIX IO
- MPI IO
- Buffered IO
- Non-Buffered IO

Parallel and Distributed Computing (CS621)

Question: 19 (Marks: 1) Attempted Questions: 43 Total Questions: 50

The time elapsed between the beginning and the end of its execution on a sequential computer is known as _____.

Choices:

- serial runtime
- parallel runtime
- distributed runtime
- both sequential and parallel runtime

Parallel and Distributed Computing (CS621)

Question: 20 (Marks: 1) Attempted Questions: 44 Total Questions: 50

Speedup S is computed by _____.

Choices:

- $S = \Theta(n / \log n)$
- $S = \Theta(n \log n)$
- $S = \Theta(\log n)$
- $S = \Theta(\log n)^n$

Parallel and Distributed Computing (CS621)

Question: 21 (Marks: 1) Attempted Questions: 44 Total Questions: 50

The phenomenon when the speedup become greater than p (number of processing elements) is known as _____ speedup.

Choices:

- linear
- superlinear
- non-linear
- supernon-linear

Parallel and Distributed Computing (CS621)

Question: 22 (Marks: 1) Attempted Questions: 46 Total Questions: 50

_____ is the product of parallel runtime and the number of processing elements used (p x T_P).

Choices:

- Cost
- Overhead
- Speedup
- Efficiency

Parallel and Distributed Computing (CS621)

Question: 23 (Marks: 1) Attempted Questions: 47 Total Questions: 50

Using fewer than the maximum possible number of processing elements to execute a parallel algorithm is called _____.

Choices:

- scaling up
- scaling down
- scaling out
- scaling in

Question: 24 (Marks: 1) Attempted Questions: 48 Total Questions: 50

_____ is a special class of distributed data-store which is characterized by the lack of simultaneous updates. Here, the emphasis is more on maintaining a consistent view of things for the individual client process that is currently operating on the data-store.

Choices:

- Sequential model
- Client centric consistency model
- Data consistency model
- Incremental model

Question: 25 (Marks: 1) Attempted Questions: 49 Total Questions: 50

_____ the data and moving it closer to where it is needed helps to solve the scalability problem.

Choices:

- Replicating
- Aborting
- Terminating
- Paradigm